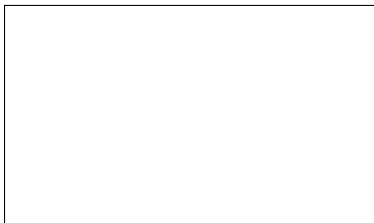


Graphical Abstract

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J.K. Krishnan, Han Thane, William J. Hansen, T. Rafeeq



Highlights

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- Research highlights item 1
- Research highlights item 2
- Research highlights item 3

This is a specimen a_b title^{*,**}

Sir J.K. Krishnan^{a,c,*,1} (Researcher), Han Thane^{b,d}, William J. Hansen Jr^{b,c,2} (Co-ordinator) and T. Rafeeq^{a,c,*,1,3}

^aDepartment of Physics, J.K. Institute of Science, Jawahar Nagar, Trivandrum, 695013, Kerala, India

^bWorld Scientific University, Street 29, 1011 NX Amsterdam, The Netherlands

^cUniversity of Intelligent Studies, Street 15, Jabaladesh, 825001, Orissa, India

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ABSTRACT

This template helps you to create a properly formatted L^AT_EX manuscript.

`\beginabstract ... \endabstract` and `\begin{keyword} ... \end{keyword}` which contain the abstract and keywords respectively.

Each keyword shall be separated by a `\sep` command.

1. Introduction

2. Numerical methodology

2.1. Overall numerical framework

A coupled numerical framework was developed to evaluate the aerodynamic response of an airfoil containing internal rectangular flow channels. The external flow was solved using a two-dimensional source–vortex panel method, while the internal flow through each channel was calculated from the pressure difference between its two surface openings. The resulting channel flow was converted into a surface-normal transpiration velocity and imposed as a modified boundary condition in the panel method. The final aerodynamic loads were obtained by combining the pressure force acting on the remaining external solid surface with the reaction force generated by wall shear inside the channels. All internal-flow quantities were evaluated per unit span.

Figure 1 summarises the numerical procedure. The flowchart is intentionally limited to the main calculation stages because all porous configurations considered in this study produced stable coupled solutions.

*This document is the results of the research project funded by the National Science Foundation.

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This note has no numbers. In this work we demonstrate a_b the formation Y_{-1} of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab.

*Corresponding author

**Principal corresponding author

 jkk@example.in (J.K. Krishnan); wjh@example.org (W. J. Hansen); t.rafeeq@example.in (T. Rafeeq)

 www.jkkrishnan.in (J.K. Krishnan); <https://www.university.org> (W. J. Hansen); www.campus.in (T. Rafeeq)

ORCID(s): 0000-0001-0000-0000 (J.K. Krishnan)

¹This is the first author footnote, but is common to third author as well.

²Another author footnote, this is a very long footnote and it should be a really long footnote. But thssdsdis footnote is not yet sufficiently long enough to make two lines of footnote text.

2.2. External-flow source–vortex panel method

2.2.1. Potential-flow formulation

The external flow was assumed to be steady, two-dimensional, incompressible, inviscid and irrotational. Under these assumptions, the velocity field can be expressed in terms of a velocity potential ϕ ,

$$\mathbf{V} = \nabla\phi, \quad (1)$$

and conservation of mass reduces to Laplace's equation,

$$\nabla^2\phi = 0. \quad (2)$$

The linearity of Eq. (2) permits the external velocity field to be represented by the superposition of the freestream, constant-strength source panels and a global vortex contribution. The source distribution represents the airfoil thickness and enforces the normal boundary condition, while the vortex strength introduces circulation and is determined from the trailing-edge Kutta condition [? ? ?].

2.2.2. Panel discretisation and boundary conditions

The closed airfoil boundary was divided into N straight panels. For panel i , the panel length was

$$S_i = \sqrt{(X_{B,i+1} - X_{B,i})^2 + (Y_{B,i+1} - Y_{B,i})^2}, \quad (3)$$

and the control point was placed at the panel midpoint. The unit tangent and outward unit normal vectors are denoted by $\mathbf{t}_i$ and $\mathbf{n}_i$, respectively. A constant source strength λ_j was assigned to every panel and a single global vortex strength γ was applied to the complete airfoil.

The normal velocity at control point i was written as

$$V_\infty \cos \beta_i + \frac{1}{2\pi} \sum_{j=1}^N I_{ij} \lambda_j - \frac{\gamma}{2\pi} \sum_{j=1}^N K_{ij} = v_{n,i}, \quad (4)$$

where I_{ij} and K_{ij} are the source and vortex normal-influence coefficients, β_i is the angle between the panel

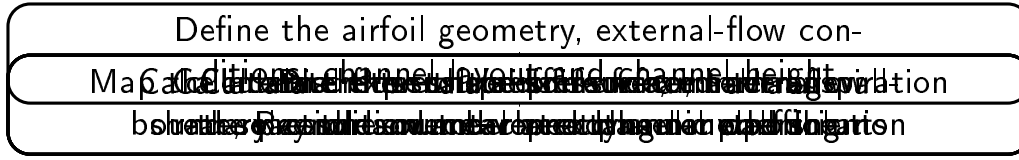


Figure 1: Numerical procedure used to couple the external source–vortex panel method with the internal rectangular-channel flow model.

normal and the freestream, and $v_{n,i}$ is the prescribed outward normal transpiration velocity. For a solid airfoil, $v_{n,i} = 0$. For a porous airfoil, $v_{n,i}$ is calculated from the internal-channel solution.

The source self-influence was treated using $I_{ii} = \pi$. Smooth flow departure at the trailing edge was imposed through the Kutta condition,

$$V_{t,1} + V_{t,N} = 0. \quad (5)$$

The unknown source and vortex strengths were assembled into

$$\mathbf{x} = [\lambda_1 \quad \lambda_2 \quad \cdots \quad \lambda_N \quad \gamma]^T, \quad (6)$$

and the aerodynamic system was written in the compact form

$$\mathbf{Ax} = \mathbf{b}. \quad (7)$$

For a fixed airfoil geometry, the influence matrix $\mathbf{A}$ remains unchanged and the porous flow modifies the first N entries of $\mathbf{b}$ through the term $2\pi v_{n,i}$. Equation (7) was solved using LU factorisation.

2.2.3. Surface velocity and pressure

After solving for λ_j and γ , the tangential velocity on panel i was evaluated as

$$V_{t,i} = V_\infty \sin \beta_i + \frac{1}{2\pi} \sum_{j=1}^N J_{ij} \lambda_j + \frac{\gamma}{2} - \frac{\gamma}{2\pi} \sum_{j=1}^N L_{ij}, \quad (8)$$

where J_{ij} and L_{ij} are the tangential source and vortex influence coefficients. At a panel containing a finite opening, the local speed used in Bernoulli's equation includes both tangential and normal components. The pressure coefficient was therefore calculated as

$$C_{p,i} = 1 - \frac{V_{t,i}^2 + \overline{v_{ni}^2}}{V_\infty^2}, \quad (9)$$

where $\overline{v_{ni}^2}$ is the panel-length-weighted mean square of the normal velocity over the open part of panel i . For a completely solid panel, $\overline{v_{ni}^2} = 0$, and Eq. (9) reduces to

the conventional panel-method expression. The dimensional surface pressure was

$$p_i = p_\infty + \frac{1}{2} \rho_\infty V_\infty^2 C_{p,i}. \quad (10)$$

2.3. Porous-airfoil internal-flow model

2.3.1. Porous-flow concept and channel geometry

The porous airfoil was represented by straight, constant-height rectangular channels embedded in the two-dimensional airfoil cross-section. Each channel connected two finite openings on the external surface and was solved independently per unit span. Three layouts were considered: four horizontal channels, four vertical channels, and an independent superposition of the horizontal and vertical networks. Intersections between the two layouts in the combined configuration were treated as independent layers and did not exchange internal flow.

For channel m , the full channel height, centreline length and unit direction vector are denoted by h_m , L_m and $\mathbf{e}_m$, respectively. The channel height is the flow area per unit span, while the hydraulic diameter of the two-dimensional slot is

$$D_{h,m} = 2h_m. \quad (11)$$

Because a rectangular channel cuts the airfoil skin over a finite length, the opening was mapped to the exact overlapping portions of the surface panels rather than to a single control point. For opening o , with local outward surface normal $\mathbf{n}_{s,o}$, the skin-intersection length was

$$\ell_o = \frac{h_m}{|\mathbf{e}_m \cdot \mathbf{n}_{s,o}|}. \quad (12)$$

Let ℓ_{io} be the overlap length between opening o and panel i . The pressure assigned to the opening was obtained from the overlap-length-weighted average

$$p_o = \frac{\sum_{i \in o} p_i \ell_{io}}{\sum_{i \in o} \ell_{io}}. \quad (13)$$

This treatment preserves the physical opening size and allows an opening to terminate within a panel without remeshing the complete airfoil boundary.

2.3.2. Internal rectangular-channel equations

For channel m , the pressure difference driving the internal flow was

$$\Delta p_m = p_{1,m} - p_{2,m}, \quad (14)$$

where a positive value denotes flow from opening 1 to opening 2. The flow rate per unit span is denoted by q_m and has units of $\text{m}^2 \text{s}^{-1}$. The mean internal velocity and Reynolds number were calculated as

$$U_m = \frac{q_m}{h_m}, \quad (15)$$

$$Re_m = \frac{\rho_\infty |U_m| D_{h,m}}{\mu} = \frac{2\rho_\infty |q_m|}{\mu}. \quad (16)$$

For $Re_m < 2300$, fully developed laminar flow between parallel walls was represented directly by the plane-Poiseuille relation [? ?],

$$q_m = \frac{\Delta p_m h_m^3}{12\mu L_m}. \quad (17)$$

For $Re_m \geq 2300$, the pressure loss was described using the Darcy–Weisbach equation,

$$|\Delta p_m| = f_m \frac{L_m}{D_{h,m}} \frac{\rho_\infty U_m^2}{2}, \quad (18)$$

with the smooth-channel Blasius approximation

$$f_m = \frac{0.3164}{Re_m^{0.25}}. \quad (19)$$

Because f_m depends on Re_m and therefore on q_m , the turbulent flow rate was obtained using a bracketed bisection solution of Eq. (18). Entrance, exit, bend and junction losses were not included; therefore, the internal pressure loss represents distributed wall-friction resistance in a straight constant-height channel.

2.3.3. Flow mapping and channel wall shear

The channel flow was converted into a normal velocity at each surface opening. For opening o , the uniform local opening velocity was

$$v_{o,m} = s_o \frac{q_m}{\ell_o}, \quad (20)$$

where $s_o = -1$ for flow entering the airfoil and $s_o = +1$ for flow leaving the airfoil. The panel-averaged contribution of this opening was

$$v_{n,i}^{(o)} = v_{o,m} \frac{\ell_{io}}{S_i}. \quad (21)$$

The contributions from all openings intersecting a panel were summed. This mapping conserves the per-unit-span flow rate because

$$\sum_{i \in o} v_{n,i}^{(o)} S_i = s_o q_m. \quad (22)$$

The mean-square normal velocity used in Eq. (9) was calculated consistently as

$$\overline{v_{ni}^2} = \sum_o \left(\frac{q_m}{\ell_o} \right)^2 \frac{\ell_{io}}{S_i}. \quad (23)$$

For laminar plane-Poiseuille flow, the magnitude of the wall shear stress was

$$\tau_{w,m} = \frac{6\mu |U_m|}{h_m}. \quad (24)$$

For the Darcy–Weisbach branch, the equivalent wall shear was calculated from

$$\tau_{w,m} = \frac{f_m}{8} \rho_\infty U_m^2. \quad (25)$$

The two channel walls produce a total resisting force per unit span of

$$F'_{w,m} = 2\tau_{w,m} L_m. \quad (26)$$

For a straight constant-height channel with only distributed wall-friction loss, the pressure and wall-shear forces satisfy

$$F'_{w,m} = |\Delta p_m| h_m, \quad (27)$$

which was used as an internal force-balance check.

2.4. Aero–hydraulic coupling

The external and internal solutions were coupled through the surface pressure and the normal transpiration velocity. The calculation was initialised using zero transpiration on all panels. The panel method then provided the surface pressure distribution, from which the two opening pressures and Δp_m were obtained for every channel. The rectangular-channel model returned the flow rate, internal Reynolds number and wall shear, and the flow-mapping procedure generated an updated normal-velocity distribution for the external solver.

A fixed-point coupling procedure was used. If $v_{n,i}^*$ denotes the normal velocity obtained directly from the current

internal-flow solution, the value supplied to the next panel solve was under-relaxed according to

$$v_{n,i}^{(k+1)} = (1 - \omega)v_{n,i}^{(k)} + \omega v_{n,i}^*, \quad (28)$$

where $\omega = 0.5$ was used in the present calculations. The same relaxation was applied to the panel distribution of $\overline{v_n^2}$. The panel and channel calculations were repeated to obtain a self-consistent pressure, internal-flow and transpiration solution. A final aerodynamic solve was then performed using the coupled boundary data before the force and moment coefficients were evaluated.

2.5. Force and moment calculation

The aerodynamic loads were divided into an external-wall contribution and an internal-channel contribution. Let w_i denote the fraction of panel i that remains a solid pressure-bearing wall. The external Cartesian force-coefficient contributions were

$$dC_{x,i}^{\text{ext}} = -C_{p,i} n_{x,i} \frac{S_i}{c} w_i, \quad dC_{y,i}^{\text{ext}} = -C_{p,i} n_{y,i} \frac{S_i}{c} w_i. \quad (29)$$

The external coefficients were obtained by summing over all panels,

$$C_x^{\text{ext}} = \sum_{i=1}^N dC_{x,i}^{\text{ext}}, \quad C_y^{\text{ext}} = \sum_{i=1}^N dC_{y,i}^{\text{ext}}. \quad (30)$$

The external lift and streamwise pressure-force coefficients were

$$C_L^{\text{ext}} = C_y^{\text{ext}} \cos \alpha - C_x^{\text{ext}} \sin \alpha, \quad (31)$$

$$C_D^{\text{ext}} = C_x^{\text{ext}} \cos \alpha + C_y^{\text{ext}} \sin \alpha. \quad (32)$$

The external pitching-moment coefficient about $(x_{\text{ref}}, y_{\text{ref}})$ was evaluated from the panel-force contributions using the sign convention adopted in the aerodynamic solver.

The channel wall applies a resisting shear force to the internal fluid. By Newton's third law, the fluid applies an equal and opposite force to the airfoil in the internal flow direction. If $\mathbf{t}_{q,m}$ is the unit vector in the solved flow direction, the force exerted on the airfoil by channel m is

$$\mathbf{F}'_{m,\text{int}} = F'_{w,m} \mathbf{t}_{q,m}. \quad (33)$$

After summing the channel forces, the internal Cartesian coefficients were

$$C_x^{\text{int}} = \frac{\sum_m F'_{x,m,\text{int}}}{\frac{1}{2} \rho_\infty V_\infty^2 c}, \quad C_y^{\text{int}} = \frac{\sum_m F'_{y,m,\text{int}}}{\frac{1}{2} \rho_\infty V_\infty^2 c}. \quad (34)$$

The internal lift coefficient was then

$$C_L^{\text{int}} = C_y^{\text{int}} \cos \alpha - C_x^{\text{int}} \sin \alpha. \quad (35)$$

The internal pitching-moment contribution was evaluated about the same reference point using the channel mid-point and the internal-force vector. The total aerodynamic coefficients were obtained by adding the two contributions,

$$C_L = C_L^{\text{ext}} + C_L^{\text{int}}, \quad C_D = C_D^{\text{ext}} + C_D^{\text{int}}, \quad C_M = C_M^{\text{ext}} + C_M^{\text{int}}. \quad (36)$$

The reported streamwise coefficient contains the inviscid external pressure-force contribution and the internal channel wall-shear contribution. It does not include external boundary-layer skin friction, separation losses or wake momentum deficit and should therefore not be interpreted as a complete viscous drag prediction.

3. Installation

4. Front matter

5. Bibliography styles

6. Floats

7. Theorem and theorem like environments

cas-dc.cls provides a few shortcuts to format theorems and theorem-like environments with ease. In all commands the options that are used with the `\newtheorem` command will work exactly in the same manner. cas-dc.cls provides three commands to format theorem or theorem-like environments:

```
\newtheorem{theorem}{Theorem}
\newtheorem{lemma}[theorem]{Lemma}
\newdefinition{rmk}{Remark}
\newproof{pf}{Proof}
\newproof{pot}{Proof of Theorem \ref{thm2}}
```

The `\newtheorem` command formats a theorem in L^AT_EX's default style with italicized font, bold font for theorem heading and theorem number at the right hand side of the theorem heading. It also optionally accepts an argument which will be printed as an extra heading in parentheses.

```
\begin{theorem}
  For system (8), consensus can be achieved with
  $\lvert T_{\omega} z \rvert \leq \dots$
  \begin{eqnarray}\label{10}
    \dots
  \end{eqnarray}
\end{theorem}
```

- Parenthetical: `\cite{WB96}` produces (Wettig & Brown, 1996).
- Textual: `\citet{ESG96}` produces Elson et al. (1996).

- In the numbered scheme of citation, `\cite{<label>}` is used, since `\citep` or `\citet` has no relevance in the numbered scheme. `natbib` package is loaded by `cas-dc` with `numbers` as default option. You can change this to `author-year` or `harvard` scheme by adding option `authoryear` in the class loading command. If you want to use more options of the `natbib` package, you can do so with the `\biboptions` command. For details of various options of the `natbib` package, please take a look at the `natbib` documentation, which is part of any standard L^AT_EX installation.

Appendix sections are coded under `\appendix`.
`\printcredits` command is used after appendix sections
to list author credit taxonomy contribution roles tagged using
`\credit` in frontmatter.

J.K. Krishnan: Conceptualization of this study, Methodology, Software. **William J. Hansen:** Data curation, Writing - Original draft preparation.

- [1] Fortunato, S., 2010. Community detection in graphs. *Phys. Rep.-Rev. Sec. Phys. Lett.* 486, 75–174.
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